

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 12

Level 1 — Beginner · Length, Weight & Capacity I

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 1 — Beginner	Topic Focus	Measurement I — Units & Conversions
Questions	20	Time Allowed	30 minutes
Total Marks	20	Worksheet No.	12 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 12 starts the Measurement block — the single biggest chapter in the SOF IMO for Class 4. The child masters the metric ladders (km-m-cm, kg-g, L-mL), converts mixed units like 2 m 40 cm into centimetres, chooses the sensible unit for real objects, and compares quantities by converting to a single unit first — the golden rule of all measurement problems.

Learning Objectives

- Recall the metric conversions: 1 km = 1,000 m, 1 m = 100 cm, 1 kg = 1,000 g, 1 L = 1,000 mL.
- Convert between units, including mixed units like 1 m 35 cm.
- Choose the most sensible unit for measuring real-life objects.
- Compare lengths, weights and capacities by converting to one unit.
- Solve simple measurement word problems with halves and equal parts.

Instructions

- This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
- Read every question carefully. Look at any picture, table or figure before choosing.
- Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
- Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
- Do all rough work in the margin or on rough paper. Do not use a calculator.
- After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

- Q1.** How many **centimetres (cm)** are there in 1 metre (m)?
(A) 10 (B) 100
(C) 1,000 (D) 50
- Q2.** How many **metres (m)** are there in 1 kilometre (km)?
(A) 100 (B) 500
(C) 1,000 (D) 10,000
- Q3.** How many **grams (g)** are there in 1 kilogram (kg)?
(A) 10 (B) 100
(C) 1,000 (D) 10,000
- Q4.** How many **millilitres (mL)** are there in 1 litre (L)?
(A) 10 (B) 100
(C) 1,000 (D) 500
- Q5.** Convert: **3 m = ___ cm**
(A) 30 (B) 300
(C) 3,000 (D) 33
- Q6.** Convert: **5 kg = ___ g**
(A) 50 (B) 500
(C) 5,000 (D) 50,000
- Q7.** Convert: **2 m 40 cm = ___ cm**
(A) 240 (B) 204
(C) 2,040 (D) 24
- Q8.** Convert: **4,000 mL = ___ L**
(A) 4 (B) 40
(C) 400 (D) 0.4
- Q9.** Which unit is BEST for measuring the length of a **pencil**?
(A) Kilometre (B) Metre
(C) Centimetre (D) Litre
- Q10.** Which unit is BEST for measuring the distance between two CITIES?
(A) Centimetre (B) Metre
(C) Kilometre (D) Gram
- Q11.** Which unit is BEST for weighing a **school bag full of books**?
(A) Gram (B) Kilogram
(C) Litre (D) Metre

Q12. Which of these would be measured in **litres**?

- (A) The weight of rice (B) The milk in a bucket
(C) The length of a rope (D) The height of a door

Q13. A ribbon is 85 cm long. Another ribbon is 1 m long. Which is **LONGER**, and by how much?

- (A) First, by 15 cm (B) Second, by 15 cm
(C) Second, by 85 cm (D) Both equal

Q14. Which is the **HEAVIEST**?

- (A) 900 g (B) 1 kg
(C) 1 kg 200 g (D) 1,100 g

Q15. A bottle holds 500 mL of juice. How many bottles fill a 2-litre jug?

- (A) 2 (B) 3
(C) 4 (D) 5

Q16. Ravi's height is 1 m 35 cm and his brother's height is 128 cm. Who is **TALLER**?

- (A) Ravi (B) His brother
(C) Both equal (D) Cannot say

Q17. Half a kilogram is the **SAME** as:

- (A) 50 g (B) 500 g
(C) 5,000 g (D) 100 g

Q18. A rope of 6 m is cut into 2 **EQUAL** pieces. How long is each piece in centimetres?

- (A) 3 cm (B) 30 cm
(C) 300 cm (D) 600 cm

Q19. Which is **MORE**: 1 L 250 mL or 1,500 mL?

- (A) 1 L 250 mL (B) 1,500 mL
(C) Both equal (D) Cannot compare

Q20. A vegetable seller weighs 250 g of chillies, 750 g of tomatoes and 1 kg of potatoes. What is the **TOTAL** weight?

- (A) 1 kg 500 g (B) 2 kg
(C) 2 kg 250 g (D) 1 kg 750 g

ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	C	C	C	B	C	A	A	C	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	B	B	B	C	C	A	B	C	B	B

Step-by-Step Solutions

Q1. Correct answer: (B)

1 metre = **100** centimetres.

Q2. Correct answer: (C)

1 kilometre = **1,000** metres.

Q3. Correct answer: (C)

1 kilogram = **1,000** grams.

Q4. Correct answer: (C)

1 litre = **1,000** millilitres.

Q5. Correct answer: (B)

1 m = 100 cm, so 3 m = $3 \times 100 =$ **300** cm.

Q6. Correct answer: (C)

1 kg = 1,000 g, so 5 kg = $5 \times 1,000 =$ **5,000** g.

Q7. Correct answer: (A)

2 m = 200 cm. Then $200 + 40 =$ **240** cm.

Q8. Correct answer: (A)

1,000 mL = 1 L, so 4,000 mL = $4,000 \div 1,000 =$ **4** L.

Q9. Correct answer: (C)

A pencil is small, so we measure it in **centimetres**. (A litre measures liquids, not length!)

Q10. Correct answer: (C)

Long distances between cities are measured in **kilometres**.

Q11. Correct answer: (B)

A full school bag is heavy (a few thousand grams), so we use **kilograms**.

Q12. Correct answer: (B)

Litres measure liquids — like **milk in a bucket**. Rice is weighed (kg), rope and door are lengths (m/cm).

Q13. Correct answer: (B)

1 m = 100 cm. $100 - 85 = 15$, so the **second ribbon is longer by 15** cm.

Q14. Correct answer: (C)

Convert all to grams: 900 g; 1,000 g; **1,200** g; 1,100 g. The heaviest is 1 kg 200 g.

Q15. Correct answer: (C)

2 L = 2,000 mL. $2,000 \div 500 =$ **4** bottles.



Q16. Correct answer: (A)

Ravi = $100 + 35 = 135$ cm, brother = 128 cm. $135 > 128$, so **Ravi** is taller.

Q17. Correct answer: (B)

Half of 1,000 g = **500 g**.

Q18. Correct answer: (C)

$6 \text{ m} \div 2 = 3 \text{ m}$, and $3 \text{ m} = \mathbf{300 \text{ cm}}$.

Q19. Correct answer: (B)

$1 \text{ L } 250 \text{ mL} = 1,250 \text{ mL}$. Since $1,500 > 1,250$, **1,500 mL** is more.

Q20. Correct answer: (B)

$250 + 750 = 1,000 \text{ g} = 1 \text{ kg}$. Plus 1 kg of potatoes = **2 kg** total.

